

Edge Detection During Image Processing Using MATLAB



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Image processing technologies have advanced at a very fast pace during the last few years. Sophisticated algorithms and methodologies that result into accurate analysis of the object make this area really special. Image processing has unlimited applications such as object detection, edge detection, framing and characterisations to name a few.

Another speciality of image processing is its compatibility with most of the widely used environments in the industry, such as C, C++, Java, MATLAB, CUDA or any other high-end object-oriented software. So image processing is possible everywhere.

Presented here is one such popular algorithm for edge detection of an object. Edge detection is the name of a set of mathematical methods that aim at identifying points in a digital image at which the image brightness changes sharply, or more formally, and has discontinuities, as defined by Wikipedia. Fig. 1 shows an image processed using Canny's algorithm for edge detection.

The platform used is MATLAB, and a very elegant graphic user interface (GUI) has been developed for ease of use and aesthetic appeal to the user or learner as shown in Fig. 2.

Canny's edge-detection algorithm

In Canny's edge-detection method, the edge detection block finds edges by looking for the local maxima of the gradient of the input image. It calculates the gradient using the derivative of the Gaussian filter. The method uses two thresholds to detect strong and weak edges. It includes the weak edges in the output only if these are connected to strong edges. As a result, the



Fig. 1: Edge detection using Canny's algorithm (Courtesy: Wikipedia)



Fig. 2: GUI edge-detection program

method is more robust to noise, and more likely to detect true weak edges.

If you want to use directly the in-built function in MATLAB, the syntax preferred is:

```
[g t] = edge (f, 'canny', T, sigma)
```

where 'T' is the vector containing the threshold values.

The algorithm actually works in five separate steps:

1. Smoothing: Blurring of the image to remove noise.
2. Finding gradients: The edges are marked where the gradients of the image have large magnitudes.
3. Non-maximum suppression: Only local maxima should be marked as edges.
4. Double thresholding: Potential edges are determined by thresholding.

5. Edge tracking by hysteresis: Final edges are determined by suppressing all the edges that are not connected to a very certain (strong) edge.

MATLAB program

MATLAB provides a GUI development platform where engineers can develop a very aesthetic graphic interface for their application. Here in this project, GUI has been developed in MATLAB to take advantage of its features. A GUI allows users to perform tasks interactively through controls such as buttons and sliders. Within MATLAB, GUI tools enable you to perform tasks such as creating and customising plots (plottool), fitting curves and surfaces (cftool), and analysing and filtering signals (sptool). You can also create